

Problem

Find the period of small oscillations of a pendulum of mass m on a string of length l , mounted on a cart of mass M . The cart is on a smooth horizontal surface.

Solution

Let O denote the equilibrium position of the pendulum. At some time after passing through equilibrium, let the pendulum have displaced by a distance x relative to the ground, and let the cart have displaced by a distance y in the opposite direction (so that $x, y > 0$ correspond to opposite directions).

Energy of the system

The total mechanical energy is:

$$E = \frac{mv^2}{2} + \frac{MV^2}{2} + mgh, \quad (1)$$

where $v \approx v_x$ is the horizontal velocity of the pendulum (the vertical component is negligible for small oscillations), V is the velocity of the cart, and h is the height the pendulum rises.

For small angles, the horizontal displacement of the pendulum relative to the cart is $x + y$. The height gain is:

$$h = \frac{(x + y)^2}{2l}, \quad \text{assuming } x + y \ll l. \quad (2)$$

Conservation laws

Since no external horizontal forces act on the system, the horizontal component of the total momentum is conserved. Initially the system is at rest, so:

$$m\dot{x} - M\dot{y} = 0 \quad \implies \quad m\dot{x} = M\dot{y}. \quad (3)$$

(The minus sign accounts for the opposite directions; y here is taken as a positive magnitude.)

Equivalently, the position of the center of mass remains fixed. Choosing the origin such that the center of mass is at zero:

$$mx - My = 0 \quad \implies \quad y = \frac{m}{M}x. \quad (4)$$

Equations (3) and (4) are equivalent (one follows from the other by integration with zero initial conditions).

Reduction to harmonic oscillator

Substitute $y = \frac{m}{M}x$ into (2) and the kinetic energy expression.

The horizontal displacement of the pendulum relative to the cart:

$$x + y = x + \frac{m}{M}x = x \left(1 + \frac{m}{M}\right) = x \frac{M + m}{M}.$$

Potential energy:

$$U = mgh = mg \frac{(x + y)^2}{2l} = \frac{mg}{2l} \left(\frac{M + m}{M}\right)^2 x^2.$$

Kinetic energy:

$$T = \frac{m\dot{x}^2}{2} + \frac{M\dot{y}^2}{2} = \frac{m\dot{x}^2}{2} + \frac{M}{2} \left(\frac{m}{M}\dot{x}\right)^2 = \frac{m\dot{x}^2}{2} + \frac{m^2}{2M}\dot{x}^2 = \frac{1}{2}m \left(1 + \frac{m}{M}\right) \dot{x}^2.$$

Total energy:

$$E = \frac{1}{2}m \left(1 + \frac{m}{M}\right) \dot{x}^2 + \frac{1}{2}mg \frac{1}{l} \left(\frac{M + m}{M}\right)^2 x^2 = \text{const.}$$

Factor out $\frac{1}{2}m \frac{M+m}{M}$:

$$E = \frac{1}{2}m \frac{M + m}{M} \left[\dot{x}^2 + \frac{g}{l} \frac{M + m}{M} x^2 \right].$$

This is of the form $\frac{1}{2}\mu\dot{x}^2 + \frac{1}{2}\mu\omega^2x^2$ with $\mu = m \frac{M+m}{M}$ and:

$$\omega^2 = \frac{g}{l} \frac{M + m}{M}.$$

Period

The period of small oscillations is:

$$T = \frac{2\pi}{\omega} = 2\pi \sqrt{\frac{l}{g} \cdot \frac{M}{M + m}}.$$

Remark

The period is smaller than that of a simple pendulum $\left(2\pi\sqrt{l/g}\right)$ because the cart moves toward the pendulum as it swings, effectively increasing the restoring force (or reducing the reduced mass).